Normalization in SQL

Normalization is a process of organizing the data into a database to :

1. Reduce data redundancy (duplicate values)
2. Improve data integrity
3. Avoid insert, update and delete anomalies

1NF (first normal form)

Rules :

Each column contains only one atomic single values

No repeating groups

Emp

Emp_id	emp_name	phone
101	smith	123,456

This is not allowed in 1nf

CORRECT WAY

EMP_ID	EMP_NAME	PHONE
101	SMITH	123
101	smith	456

2NF

2NF OR SECOND NORMAL FROM

RULES

MUST BE IN 1 NF

No partial dependency (non-key columns should depend on the whole primary key)

Ex:

Emp_id, proj_id, emp_name

If primary key =(emp_id,proj_id) , then emp_name depends only on emp_id

Not allowed

Correct way

Emp _id emp_name

Emp_project

Emp_id proj_id

** partial dependency : it means that the one column should be dependent on each other

-3NF

Third normal from

Rules

Must be in 2NF

No transitive dependency

** transitive dependency :

It means and it occurs when a non-key column depends on another non-key column instead of depending directly on primary key

Ex :

Primary key – column a – column b

Column b here depends on column a , not directly on the primary key

Ex :

Emp

Emp_id	emp_name	dept_id	dept_name
101	amith	10	it

Primary key = emp_id

Dependencies

Emp_id -dept _id

Dept_id – dept_name

Hence , emp_id – dept_id-dept_name

Dept_name here is transistevely dependent on emp_id

What is the problem

If the dept name changes

It must be update every employee row in dept this creates redundancy and problems

There should not be two primary key columns in one table

This is also transitive dependency

4NF (boyce-codd normal form)

BCNF

A 4NF or BCNF is a database normalization form

What is BCNF :

A table in BCNF follows when :

For every functional dependency X-Y,X must be a superkey

In simple words :

- If one column in a table (or set of columns) determines another column
- Than that determining column must uniquely identify each row

Ex :

Student

Student	course	instructor
Alice	DBMS	Rajeev
Archit	data science	mudit

Suppose :

- A course has only one instructor
- Functional dependency : course – instructor

The candidate key here is (Student ,course)

Here , course determines instructor but course is not a superkey

There fore , table violates BCNF

Correct way:

BCNF decomposition

Course instructor

Course	instructor
Dbms	mudit

Student course

Student	course
Alice	dbms

BCNF decomposition this satisfies rule of BCNF

** FUNCTIONAL DEPENDENCY

It means that knowing one attribute helps you to determine other attribute it is a functional dependency

** candidate key or super key : it means that this is the column in the table on basis of which you can identify other columns

Functional dependency is there then create a new table with attributes to identify that dependency or column as a super key